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Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They also measure effects on development of cancer stem-like cells (cancer initiating cells). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified and that specific mRNA expression profiles for HZE irradiation exist. In addition, we have found that transgenic mouse models of lung cancer have increased lung cancer risk particularly after fractionated high LET exposure allowing us to study the early onset and progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. HZE Particle Exposure and the Risk for Human Lung Carcinogenesis; 2. Mouse Models of Lung Cancer after HZE Particle Irradiation; 3. Organotypic (3D) Human Models of Genomic Instability for HZE Particle Induced Lung Carcinogenesis; 4. Integrating Biomarkers of Lung Cancer Pathogenesis for Risk Assessment after HZE Particle Irradiation. These projects are supported by 2 Cores: Administrative Core and the Database, Biostatistics and Bioinformatics Core. The HZE particle irradiation is done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, and a Center for the NCI Early Detection Research Network (EDRN) program, providing great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in life time never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. In addition, in the past decade in the USA, Europe, and Asia there has been a resurgent interest in radiographic screening for lung cancer, particularly using serial computed tomography (CT) scans. In fact, the USA NCI has completed a large (50,000 person) National Lung Cancer Screening Trial (NLST) that randomized persons to have CT scans or not for the early detection of lung cancer. This trial has found a significant survival benefit for persons receiving CT scan screening. This raises a variety of questions concerning the use of this approach because of its cost and substantial population based radiation exposure. Greater than 25% of persons screened (current or former smokers) will have some lesions requiring further CT scan follow-up, when biopsies are performed only 1% of the overall group will have histologic evidence of lung cancer. In addition, many of the persons will require several CT scans in follow-up to determine if the lesion is growing to warrant a biopsy. Thus, there is considerable interest in knowing the contribution of CT scan doses of gamma radiation to the pathogenesis of lung cancer. This is particularly true since this radiation is being administered to individuals (current and former smokers) who are at highest risk of having preneoplastic lesions leading to lung cancer. Our studies of low dose radiation of the preclinical human and mouse model systems will allow us to provide some risk assessment for the effect of such radiation doses (low dose) on the genesis of lung cancer and provides the rationale for DOE support of our studies. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 50 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as expression of oncogenic K-RASv12, knockdown of expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high-LET and low dose gamma irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung cancer as part of long-term missions in space. In addition, as stated we will be developing risk estimates following low dose gamma radiation as seen on Earth, particularly from CT scan studies. Thus, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through CT scan screening studies, and environmental exposures, accidentally or through terrorism. A part of this will be to understand the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional (3D) cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic,

epigenetic, mRNA and protein expression, and signaling pathway changes. Our large HBEC panel coupled with new high density single nucleotide polymorphism (SNP) analysis will allow describing polymorphic differences in such responses. Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions: "We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models ; "We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate? ; "We are developing tissue specific risk models using human 3D culture and animal models for lung cancer ; "We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, altered senescence, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated.
Overall we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. As will be described in this report, we are confident and will provide compelling evidence that the sum of the efforts of our assembled team are significantly contributing more to assessing cancer risk associated with space radiation compared to the individual components.

Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to Project 1 HZE Particle Exposure and the Risk for Human Lung Carcinogenesis (Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation). Michael Story, Ph.D. Lead, Adi F. Gazdar, M.D. Co-Lead, John D. Minna, M.D. Co-Lead
Specific Aim 1. Identify the radiation parameters, (dose, dose rate, Z, energy, LET) that drive radiation-induced cellular transformation to malignancy in normal HBEC/HSAECs.

Specific Aim 2. Determine the role of gender, lung cell of origin, and inter-individual variation in HZE particle-induced lung carcinogenesis using a large cohort of normal HBEC/HSAEC lines derived from more than 50 individuals.

Specific Aim 3. Determine the nature and frequency of genetic and epigenetic changes, and pulmonary stem cell markers associated with HZE particle-induced lung cancer.

Role of Z and energy in eliciting the maximum biologic response (maximal RBE): A major goal of this project was to determine the radiosensitivity of normal HBECS to particles of increasing Z and energy in order to determine the LET that elicits the maximum biological response. The results to date show the RBE for fraction of cell survival at a value of 0.38. Interestingly, we find the maximal RBE is at or near 150 keV/u. As a result, other combinations of Z and energy are necessary to determine where the maximal LET is actually achieved. Role of human bronchial epithelial cell transformation and radiosensitivity after HZE particle radiation: In the last report we described our results for cellular transformation (growth in soft agar) after HZE exposure and the initial characterization of isolated transformed cells. A significant feature of those results was the finding that in HBECs that expressed a mutant RAS and exhibited down-regulation of p53, (HBEC3KT R53) there was a high degree of epithelial to mesenchymal transition (EMT). This was not seen in the isogenic parental HBEC3-KT cell line. There appears to be an interesting trend in radiosensitivity associated with the various transformed cell lines collected. Besides the parental HBEC3-KT cell line (3KT) there are two groups, A and B. Group A represents transformed cell lines isolated from unirradiated HBEC3-KT cells, while group B represents transformed cell lines isolated from irradiated HBEC3-KT cells, (g-, Si or Fe). Unexpectedly, those clones isolated from irradiated HBEC3-KT cells are more radiosensitive compared to clones isolated from unirradiated HBEC3-KT cells. Contrast this finding with cellular survival for the HBEC3-KTR53 cell line and respective transformed clones are compared. Here, transformed clones that have undergone EMT after radiation are compared to transformed clones after radiation where this transition has not occurred. It would appear that cells that have undergone EMT are more radioresistant while those that have not undergone EMT are comparable to the untransformed parental HBEC3-KTR53. This result suggests that cells that are more progressed down the pathway leading to lung cancer, when exposed to radiation, have a higher likelihood of becoming radioresistant. Molecular and Karyotypic Changes in HBECs undergoing transformation after HZE Particle Radiation: We have found variable changes across cell lines, however, there are three consistent findings for the HBEC3-KT cell lines: up-regulation of MDM2; up-regulation of CDK6; and no evidence for mutations in K-, H- or N-RAS. Another feature of transformed cell lines originating from both HBEC3-KT and HBEC3-KTR53 cells is aneuploidy.

Project 2: Mouse Models of Lung Cancer after HZE Particle Irradiation. Jerry W. Shay, Ph.D. Project Lead

Using transgenic mouse models of lung cancer this project will determine the role of inflammatory cell responses in murine lung cancer arising after HZE particle exposure. Using genome wide technology, we are comparing the genetic changes in radiation induced versus. spontaneous arising murine lung cancers. All of these results and unique bioreagents will be made available to other NASA Lung Cancer Consortium Members.

Aim 1: To test the hypothesis that the inflammatory signatures are biomarkers of fractionated high LET induced invasive lung cancer.

Aim 2: To test if single dose acute but not fractionated equivalent dose high LET IR leads to increased stem cell killing. We will determine if fractionated high LET IR but not single acute doses of IR "wake up" quiescent stem cells leading to cancer progression.

Aim 3: To determine the molecular events (using genome-wide technologies) occurring in murine HZE particle IR associated lung cancers compared to mouse lung cancer arising in the absence of HZE IR. We will use our current mouse model and compare to other mouse models of lung cancer as are developed by other NASA Lung Cancer Consortium members.

Molecular studies on lung cancer progression in humans have revealed that one of the hallmark changes includes oncogenic K-ras mutations. Expression of oncogenic K-ras in murine lung initiates the formation of lesions that mimic lung cancer progression in humans and permits evaluation of lung cancer progression in vivo after exposure to varying types, doses and dose rates of irradiation. We have chosen to use the LA-1 K-ras lung cancer susceptible mouse model to examine the cancer promoting effects of heavy ions such as Si and Fe as well as solar particle events (SPE) simulations. The long term goal of the study is assess the risk to lethal lung cancers and to then extrapolate the data to human risk projections. A key NASA objective is to extrapolate from cell and animal models to human risk and our

Task Progress:

initial results suggest that this will be possible. The murine model (LA-1) randomly expresses mutated K-ras in a subset of lung cells resulting in initiation and formation of lesions which mimic lung cancer progression in humans. Greater than 50% of the mice with oncogenic K-ras expression die in less than a year with a small percent of mutant mice living to a maximum of about 600 days approximately 200-300 day less than wild-type mice. About~15% of LA-1 mice spontaneously develop invasive (lethal) non small cell lung carcinomas after about one year. In previous studies we sought to determine if X-ray or charged particle radiation exposure affects initiation or progression of lung cancer in mouse models. We used whole body irradiation (as an acute 1.0 Gy dose or a fractionated dose of 0.2 Gy per day for 5 days) of an oncogenic K-ras mouse model (LA1), which is predisposed to benign tumors. We observed that irradiation did not impact the number or size of benign lung tumors initiated in the animals. However, approximately one year after irradiation, histopathological analysis of tumors from LA1 K-ras mutant mice irradiated with the fractionated 1.0 Gy dose of 56Fe particles (1GeV) demonstrated a dramatic progression in grade of lesion compared to all other groups. Importantly, LA1 K-ras mice that received fractionated doses of 56Fe particles (but not an acute dose) showed a significant increase in the number of invasive adenocarcinomas.

Due to an increasing understanding of the role of the tumor microenvironment and inflammation on lung cancer progression, experiments were initiated to determine if signatures of advanced disease could be detected significantly earlier than one year after irradiation. Thus, we repeated the above experiments and isolated lung tissue 70 days after irradiation (a time point that where only low grade hyperplastic lesions are present and over 200 days prior to the histological observation of advanced invasive tumors). We then conducted genome wide expression analysis of whole lung (15 mice each of unirradiated, fractionated, and acute 56Fe particles irradiation) and analyzed the results by hierarchal clustering and principal component analysis. Based on these results, we then sought to extrapolate from experimental mouse data to human archival lung cancer expression datasets. After matching genes between mouse and human array platforms we then compared the signatures from our mouse data to determine if there was a correlation with overall human lung cancer survival. Based on a small gene classifier set, we were able to show across several independent data sets that the irradiation mouse lung signatures had high prognostic value when applied to human lung tumors for determining the survival outcome of human patients with lung cancer. Using integration of biological networks and gene expression data (Cytoscape) we were able to identify distinct inflammatory and cell cycle responses after irradiation but prior to histopathological detection of invasive lesions. Additional validation of the proteins and pathways involved are in progress. In summary, these results suggest that sustained exposure to space radiation in the presence of pre-malignant initiated lesions could affect astronaut cancer risk in future long-term missions.

In recent experiments we administrated whole body proton irradiation, 2.0 Gy over 2 hours with a wide range of energies (50 MeV-300MeV). Initial results from the solar particle event (SPE) simulations demonstrate survival differences between irradiated and control LA-1 mice. Further analysis of the irradiated LA-1 animals demonstrated a dramatic progression in tumor grade by histopathological analysis. While the number of initiated lesions between the irradiated and unirradiated mutant were similar, mice exposed to a 2.0 Gy protons (SPE simulation) demonstrated a significant increase in occurrence of invasive adenocarcinomas. These advanced invasive lesions observed in animals exposed to a simulated solar particle event closely resemble adenocarcinomas clinically observed in humans. These results are preliminary (still small numbers of mice).

Project 3. Human Organotypic (3D) Genomic Instability Model for HZE Particle-Induced Lung Carcinogenesis. David Chen, Ph.D. Project Lead, Aroumougame Asaithamby, Ph.D. Co-Lead.

Overall Objective: With the ultimate objective of estimating cancer risks to humans from HZE particles in space flight, in this project we propose to understand the DNA damage responses of human lung epithelial cells to HZE particles, since the failure of these responses to correctly repair radiation damaged DNA and increased cancer predisposition is established. In both pre-cancerous lesions and cancers, activated oncogenes induce stalling and collapse of DNA replication forks, which in turn lead to the formation of DNA DSBs. This continuous formation of DNA DSBs may contribute to the genomic instability that characterizes the vast majority of human cancers. In what could be seen as an analogous process, the complex DSBs along the densely ionizing track induced by HZE particles are difficult to repair and, in some instances, are irreparable. These unrepaired or misrepaired DNA lesions likely contribute to the higher relative biological effectiveness for cell killing, chromosomal aberrations, mutagenesis, and carcinogenesis seen in HZE particle-irradiated cells. While the DNA damage response machinery may play other important roles in tumorigenesis, there are insufficient data to determine the relationship between early biological effects of radiation, (e.g. unrepaired and/or mis-repaired DNA lesions), and the probability of cancer in humans. It is this missing knowledge that in part leads to large uncertainties in projecting cancer risks associated with occupational or environmental (space) radiation exposures. As we are specifically interested in risks related to lung cancer development upon exposure to space radiation, we have developed two novel approaches. We have: 1) made human bronchial epithelial cells (HBECs) stably expressing EGFP-53BP1 which allows real time, and continuous monitoring of the occurrence of 53BP1 foci (a hallmark of DSBs) and their persistence over time (indicating lack of or mis-repair); and established a novel organotypic (3D) human lung epithelial cell culture model where HBEC and human small airway epithelial cells (HSAECs) grown in Matrigel overlying a lawn of lung stromal fibroblast cells differentiate into tissue-specific structures mimicking those found in normal lung. Using EGFP-53BP1 monitoring and the 3D organotypic model we have found that cells in 3D culture (compared to the same cells growing in standard “2D” monolayer tissue culture) exhibit larger numbers of unrepaired DNA lesions (EGFP-53BP1 foci) following 1 Gy 56Fe-ion irradiation (~35 vs. ~20% at 72 hrs after IR). Based on these findings, we hypothesize that unrepaired DNA lesions in HZE particle-irradiated tissues generate a ”signal” that provides selection pressure for the abrogation of DNA repair and checkpoint mechanisms and changes leading to genome instability and increased rates of cell transformation (soft agar colony formation) and full fledged tumorigenesis (in vivo in xenograft models). To test this hypothesis, we will utilize our novel human lung epithelial cell models to: a) Demonstrate that the HZE particle induced persistent DNA damage and genomic instability (by monitoring of 53BP1 foci) in human organotypic lung respiratory cell models is dependent on the dose, dose rate, energy, “Z” number and frequency (acute versus fractionated) of HZE particle radiation; b) Determine the mechanisms of HZE particles induced human lung carcinogenesis in the organotypic model. The data obtained from this aim will provide biological markers correlated to lung carcinogenesis for the development of risk assessment and radiation protection for astronauts during future space missions.

Specific Objectives

Aim 1: Determine if HZE particle-induced genome instability in organotypic lung (HBEC/HSAEC) models is dependent on the dose, dose rate, energy, “Z” number and the frequency (acute versus fractionated) of HZE particle radiation.

Aim 2: Determine the mechanisms of HZE particle-induced human lung carcinogenesis in the organotypic

HBEC/HSAEC model.

Sub Aim 2.1: To determine whether unrepaired DNA lesions generated by HZE particles constitutively activate DNA damage signaling.

Sub Aim 2.2: To determine the effects of HZE particles on the differentiation capacity of HBEC/HSAECs.

Sub Aim. 2.3: To determine the effects of HZE particles on proliferation, viability and replicative senescence of organotypic HBEC/HSAECs.

Sub Aim. 2.4: To test the role of the microenvironment on HZE particle-induced lung carcinogenesis.

Sub Aim 2.5: To determine the effect of HZE particles on the genetic or epigenetic alterations and transforming potential of organotypic HBEC/HSAEC cultures.

Summary of Current Progress: We found one answer for the inability of HBECs in organotypic culture to repair complex DNA Double-strand breaks induced Iron particles. Our data on the ingenuity pathway analysis of global gene expression profile indicated that expression of genes that have been shown to facilitate multiple DNA repair pathways were significantly down-regulated in the organotypic culture as compared with 2D culture. Some specific DSB repair proteins, such as Chek1, Ku, H2AX and BRCA1 are known to be down-regulated in terminally differentiated cells. DNA lesions induced by high-LET radiation may vary in terms of their overhang configuration and end compatibility; therefore various repair factors are required to process each specific DNA lesion type within the complex DNA damage. Thus, our results imply that the down-regulation of multiple DNA repair pathway genes in differentiated human lung epithelial cells is the molecular reason for the defective complex DNA lesions repair.

Project 4. Integrating Biomarkers for Mechanistic Understanding of Lung Cancer Pathogenesis and Risk Assessment after HZE Particle Irradiation. Yang Xie, Ph.D. (Project Lead), John D. Minna, M.D. (Project Co-Lead) Adi F. Gazdar, M.D. (Project Co-Lead)

The overall goal of our NSCOR is to provide the information required to develop a rational scientific basis for estimation of risks and mechanistic understanding for lung cancer occurring after exposure to radiation during space flight. This goal will be achieved by generating data from human lung epithelial cell and mouse models of radiation induced lung cancer pathogenesis and DNA damage (Projects 1-3). By integrating these findings using molecular analyses and functional studies, combined with systems biology approaches as well as our large legacy datasets, this project will generate a comprehensive model for lung cancer risk assessment after HZE particle irradiation. In addition, we will deliver a molecular toolkit for risk assessment of individual astronauts after space flights that will reduce the dependency on epidemiological data and population-based risk assessment and provide molecular tools of future aid to astronauts including means for early diagnosis and prevention of lung cancer.

Specific Aims:

Aim 1: Identify molecular signatures of HZE particle induced human and mouse lung epithelial cell transformation and compare these to molecular events of terrestrial developed lung cancers in never and ever smokers.

Sub Aim 1.1: Systematically explore genome-wide genetic and epigenetic changes to identify genes associated with HZE particle induced human lung epithelial cell transformation to malignancy.

Sub Aim 1.2: Identify genes and pathways that are unique to HZE associated lung cancer by systematically comparing genes associated with HZE particle induced lung cancer in human and mouse models and genes associated with terrestrial lung cancer.

Aim 2: Use computational biology and statistical approaches to identify “hub” and “master regulators” of tumorigenesis associated with HZE particle irradiation. This aim makes heavy use of the Database and biostatistical core of this NSCOR.

Aim 3: Develop and validate a risk assessment model for HZE associated lung cancer. Perform functional tests of selected genes through exogenous expression and siRNA, shRNA mediated knockdown approaches to identify genes whose changes in expression regulate the growth of HZE particle induced lung epithelial cell transformation.

Aim 4. Integrate findings in this NSCOR with those of the other NASA Lung Cancer Consortium members.

Core A: Administrative Core. Leads: John Minna and Jerry Shay, and UTSW NSCOR Administrator: Brenda Zielke. The Administrative Core has established accounts and control over disbursement of all funds from this NSCOR application. It has also kept track of all personnel employed by this NSCOR. It has provided a central place for communication and coordinated data sharing In addition, it has established a series of biweekly to monthly scientific and planning meeting of the NSCOR investigators at UTSW. Approximately 25 scientific personnel attend each of these meetings. At each meeting administrative matters are discussed, experiments are planned, administrative changes made by the senior leadership team, and most importantly a scientific presentation by one of the NSCOR investigators is made and critiqued by the other investigators. Dr. Minna serves as the Administrative Core Lead with Dr. Jerry Shay as the Administrative Core Co-Lead. The Administrative Core coordinated all of the arrangements for the participation of several NSCOR investigators for a series of NSCOR mandated meetings in 2010, and the Administrative Core coordinated travel of multiple personnel to BNL for experimental work.

NSCOR Bioinformatics, Biostatistics and Database Core (Y. Xie, Core Lead, J. Xie, Core Co-Lead; L. Girard Core Co-Lead). The research proposed by the NSCOR encompasses a broad range of activities, including studies in cell lines and animal models utilizing many different types of radiation experiments. The Bioinformatics, Biostatistics and Database Core (Core B) has the expertise to effectively and critically analyze all of these types of data. This Core assists Project investigators with hypothesis refinement, experimental design, database development and management, quality control, analysis and informative presentation of results across all projects of the NSCOR. Members of the Core participate in monthly NSCOR meetings to ensure that proper consideration of biostatistics and data management are considered prospectively during all phases of NSCOR experiments.

Specific Aims:

Aim 1: To provide valid statistical designs of experiments and conduct innovative statistical modeling, simulations, and data analyses.

	Aim 2: Data integration and quality control. Aim 3: To expand the current UTSW-NSCOR database to an ontology-driven and internet accessible database to allow data sharing and integrative analyses. As we progress on all the NSCOR projects, significant amount of new data have been generated since our 2010 annual progress report. As a result of Core B effort, a comprehensive data management system has been established that allows efficient access and sharing of all NSCOR data.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Abstracts for Journals and Proceedings	Ding L-H, Park S, Story MD. "Abrogation of miR-15/107 miRNA Group Response to Ionizing Radiation Results in Genomic Instability in Normal Human Bronchial Epithelial Cells." 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7099.pdf , Sep-2011
Abstracts for Journals and Proceedings	Story M, Ding L-H, Park S, Minna JD. "HZE Particle Radiation-Induced Gene Expression Changes Specific to Particle Type in Human Bronchial Epithelial Cells and Promoted Anchorage Independent Growth in Synergy with Oncogenic KRAS and p53 Suppression." 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7102.pdf , Sep-2011
Abstracts for Journals and Proceedings	Park S, Larsen J, Das A, Minna J, Story M. "Identification of cellular and molecular alterations associated with the transformation of human bronchial epithelial cells after HZE and g-irradiation." Invited speaker. Scientific and Policy Challenges of Particle Radiations in Medical Therapy and Space Missions. 47th Annual Meeting of the National Council of Radiation Protection & Measurements, Bethesda, MD, March 7-8, 2011. Scientific and Policy Challenges of Particle Radiations in Medical Therapy and Space Missions. 47th Annual Meeting of the National Council of Radiation Protection & Measurements, Bethesda, MD, March 7-8, 2011. , Mar-2011
Abstracts for Journals and Proceedings	Story M, Ding L-H, Park S. "The role of the miRNA 15/107 family in the response of normal human bronchial epithelial cells to low and high LET radiation exposure." Invited speaker. International Symposium for Radiation Research and Medical Physics, Fudan University, Shanghai, China May 30 – June 2, 2011. International Symposium for Radiation Research and Medical Physics, Fudan University, Shanghai, China May 30 – June 2, 2011. , May-2011
Abstracts for Journals and Proceedings	Story M, Ding L-H, Park S, Minna J. "Heavy particle biology." Invited Speaker. HZE symposium on high LET radiation in space and in therapy, Institute of Modern Physics, Chinese Academy of Sciences, Lanzhou, China May 26-27, 2011. HZE symposium on high LET radiation in space and in therapy, Institute of Modern Physics, Chinese Academy of Sciences, Lanzhou, China May 26-27, 2011. , May-2011
Abstracts for Journals and Proceedings	Delgado O, Batten K, Kaisani A, Richardson J, Xie X-J, Girard L, Wistuba I, Minna JD, Shay JW. "56Fe Irradiation Gene Expression Signatures in a Lung Cancer Susceptible Mouse Model Predicts Human Lung Cancer Outcomes." 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7027.pdf , Sep-2011
Abstracts for Journals and Proceedings	Kaisani A, Kim SB, Delgado O, Batten K, Minna JD, Shay JW. "Charged Particle Irradiation in Lung Cancer and Colon Cancer Models." 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. , Sep-2011
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "HZE particles induced clustered DNA lesions in human cells: Detection and biological consequences." 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7009.pdf , Sep-2011
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "HZE particles induced complex DNA double-strand break repair in organotypic human lung epithelial cell culture: Efficiency and biological implications." 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Meeting, League City, TX, Sept 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7010.pdf , Sep-2011
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Induction and repair of clustered DNA lesions repair in three-dimensional cell culture." Presented at Darmstadt University of Technology, Radiation Biology and DNA Repair, Darmstadt, Germany, August 2011. Darmstadt University of Technology, Radiation Biology and DNA Repair, Darmstadt, Germany, August 2011. , Aug-2011
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Visualization of Spatio-temporal Dynamics of Ionizing Radiation Induced Clustered DNA Lesions." Presented at the 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. p. 53. , Sep-2011
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Unrepaired clustered DNA lesions induce chromosome breakage in human cells." Presented at the International Symposium for Radiation Research and Medical Physics, Fudan University, Shanghai, China May 30 – June 2, 2011. International Symposium for Radiation Research and Medical Physics, Fudan University, Shanghai, China May 30 – June 2, 2011. , May-2011

Articles in Peer-reviewed Journals	Asaithamby A, Hu B, Chen DJ. "Unrepaired clustered DNA lesions induce chromosome breakage in human cells." Proc Natl Acad Sci U S A. 2011 May 17;108(20):8293-8. PubMed PMID: 21527720 , May-2011
Articles in Peer-reviewed Journals	Asaithamby A, Hu B, Delgado O, Ding LH, Story MD, Minna JD, Shay JW, Chen DJ. "Irreparable complex DNA double-strand breaks induce chromosome breakage in organotypic three-dimensional human lung epithelial cell culture." Nucleic Acids Res. 2011 Jul;39(13):5474-88. Epub 2011 Mar 18. PubMed PMID: 21421565 , Jul-2011
Articles in Peer-reviewed Journals	Chen M, Xie Y, Story M. "A new background correction method based on gamma distributed noises for Illumina Beadarray data." Commun Stat Theory Methods. 2011 Sep 1;40(17):3055-69. PubMed PMID: 21769162 , Sep-2011
Articles in Peer-reviewed Journals	Allen J, Xie Y, Girard L, Xiao G. "Comparing Statistical Methods for Constructing Large Scale Gene Networks." PLoS One (pending acceptance). November 2011. , Nov-2011
Articles in Peer-reviewed Journals	Mangala LS, Zhang Y, He Z, Emami K, Ramesh GT, Story M, Rohde LH, Wu H. "Effects of simulated microgravity on expression profile of microRNA in human lymphoblastoid cells." J Biol Chem. 2011 Sep 16;286(37):32483-90. Epub 2011 Jul 20. PubMed PMID: 21775437 , Sep-2011
Articles in Peer-reviewed Journals	Sunaga N, Imai H, Shimizu K, Shames DS, Kakegawa S, Girard L, Sato M, Kaira K, Ishizuka T, Gazdar AF, Minna JD, Mori M. "Oncogenic KRAS-induced interleukin-8 overexpression promotes cell growth and migration and contributes to aggressive phenotypes of non-small cell lung cancer." Int J Cancer. 2011 May 4. [Epub ahead of print] PubMed PMID: 21544811 ; http://dx.doi.org/10.1002/ijc.26164 published online Aug 3, 2011. , Aug-2011
Articles in Peer-reviewed Journals	Larsen JE, Minna JD. "Molecular biology of lung cancer: clinical implications." Clin Chest Med. 2011 Dec;32(4):703-40. Epub 2011 Oct 7. PubMed PMID: 22054881 , Dec-2011
Articles in Peer-reviewed Journals	Eskiocak U, Kim SB, Ly P, Roig AI, Biglione S, Komurov K, Cornelius C, Wright WE, White MA, Shay JW. "Functional parsing of driver mutations in the colorectal cancer genome reveals numerous suppressors of anchorage-independent growth." Cancer Res. 2011 Jul 1;71(13):4359-65. Epub 2011 Apr 28. PubMed PMID: 2152755 , Jul-2011
Articles in Peer-reviewed Journals	Shay JW, Jacob P, Story MD. "Lung stem cell biology in relation to carcinogenic radiation risk." ICRP (in press, November 2011). , Nov-2011
Articles in Peer-reviewed Journals	Shay JW, Cucinotta FA, Sulzman FM, Coleman CN, Minna JD. "From Mice and Men to Earth and Space: Joint NASA-NCI Workshop on Lung Cancer Risk Resulting from Space and Terrestrial Radiation." Cancer Res. 2011 Nov 5;71(22):6926-9. Epub 2011 Sep 7. PubMed PMID: 21900398 , Nov-2011